Exp 2 · Zener diode as a voltage regulator: line and load regulation

Manual pp. 9–12 · LTspice / Virtual Lab · CO3 · the manual's printed data matches the simulation below digit-for-digit

What you already have: from Exp 1, in breakdown V_Z stays ≈ 6.2 V while I_Z changes. From C&S, voltage dividers and KCL.

Three terms first:

- **Regulator:** a circuit that keeps V_o constant.
- **Line regulation:** how much V_o moves when the **input** V_{in} changes. The "line" is the supply line.
- **Load regulation:** how much V_o moves when the **load current** changes, i.e. when R_L changes.

MANUAL EXP 2, AIM

To use a zener diode for line and load regulation. Viva form: "Input rises from 14 V to 17 V and V_o rises by only 0.08 V. Where did the other 2.92 V go?"

GUESS FIRST

Answer before reading on. Hint: there are only two other parts, R1 and the zener.

How a zener regulates

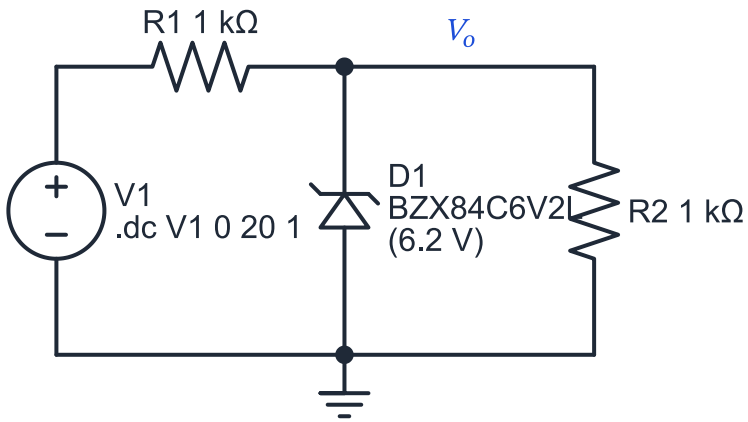


Fig 2.1 (line regulation): V1 swept 0→20 V, with a fixed 1 kΩ load.

KCL at the output node: the current through R1 splits between the zener and the load.

$$I_{R1} = \frac{V_{in} - V_Z}{R_1} = I_Z + I_L$$

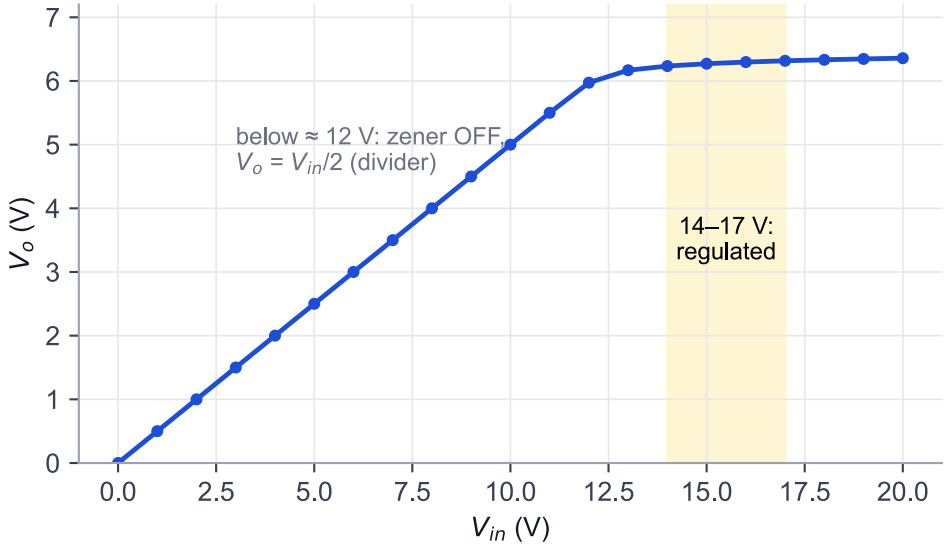
- **V_{in} rises:** the extra current all goes into the **zener**. V_Z barely moves (small r_z), so V_o stays put and **R1 absorbs the extra voltage**. That is the answer to the question.
- **R_L falls (more load current):** the zener gives up current to the load. V_o holds **as long as I_Z stays above $I_{Z(min)}$**.
- If the load demands more than R1 can supply, I_Z hits zero, the zener switches off, and V_o becomes an ordinary divider output that collapses.

Because the zener sits in parallel (shunt) with the load, this is called a **shunt regulator**.

Line regulation (Fig 2.1)

DO IT (LTSPICE)

1. Draw V1, R1 = 1k, zener **BZX84C6V2L** (cathode to the output node), R2 = 1k load, and ground.
2. Label the output wire `vo` (F4).
3. Directive: `.dc V1 0 20 1`. Run.
4. Plot V(vo) against V1. Use cursors to read V_o at V_{in} = 14 V and 17 V.



Two regions. Below about 12 V the zener is off and the output is half the input (1 kΩ:1 kΩ divider). Above it, V_o is pinned near 6.2–6.36 V.

Why 12 V? With the zener off, $V_o = V_{in}/2$. The zener turns on when $V_{in}/2$ reaches about 6.2 V, i.e. at $V_{in} \approx 12.4$ V.

V_{in} (V)	10	11	12	13	14	15	16	17	18	20
V_o (V)	5.000	5.500	5.973	6.168	**6.234**	6.271	6.297	**6.317**	6.333	6.358

$$\% \text{ line regulation} = \frac{\Delta V_o}{\Delta V_{in}} \times 100 = \frac{6.317 - 6.234}{17 - 14} \times 100 = \frac{0.083}{3} \times 100 = 2.8\,\%$$

Load regulation (Fig 2.2)

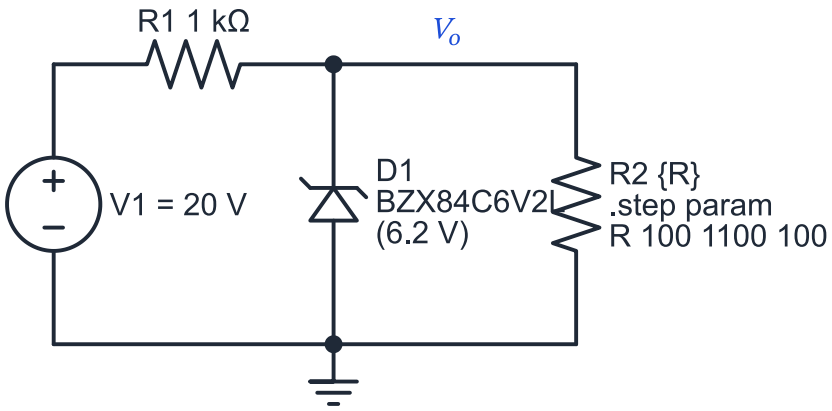
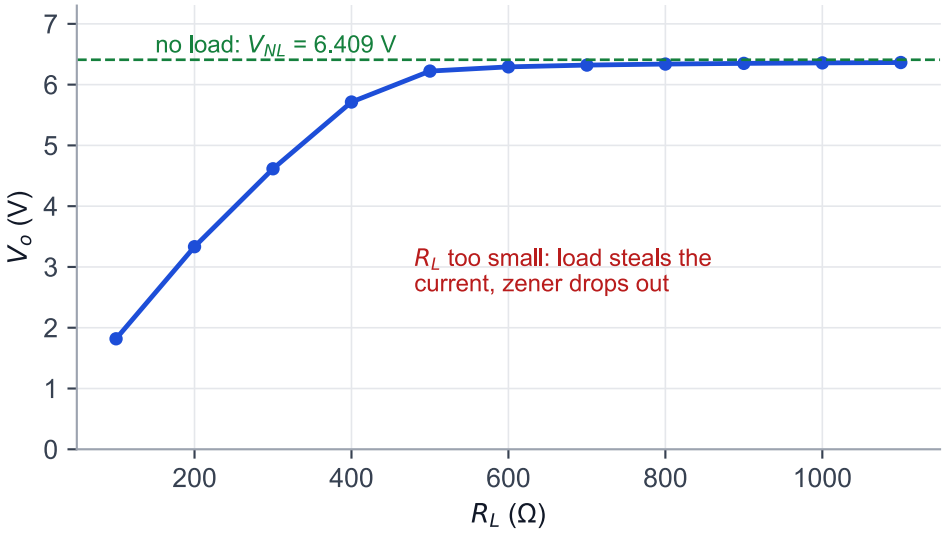


Fig 2.2 (load regulation): V1 fixed at 20 V, load stepped 100 Ω → 1100 Ω.

DO IT (LTSPICE)

1. Same circuit, but V1 = 20 V and the load value written as `{R}` (curly braces = a parameter).
2. Directives: `.step param R 100 1100 100` and `.op`. Run.
3. Plot V(vo). The x-axis is now R.
4. **No load:** delete R2, run `.op` again, and read V_{NL}.
5. Choose a full-load point (e.g. R = 1 kΩ) and read V_{FL}.



Above about 450 Ω the output holds near 6.3 V. Below that, the load takes more current than R1 can supply.

R_L (Ω)	100	200	300	400	500	600	800	1000	1100	no load
V_o (V)	1.818	3.333	4.615	5.714	6.223	6.292	6.338	**6.358**	6.364	**6.409**

$$\% \text{ load regulation} = \frac{V_{NL} - V_{FL}}{V_{FL}} \times 100 = \frac{6.409 - 6.358}{6.358} \times 100 = 0.81\ \%$$

Smallest load that still regulates: R1 can pass at most (20 − 6.2)/1 kΩ = 13.8 mA. The load gets all of that only when $I_Z \to 0$, so $R_{L(min)}$ = 6.2 V / 13.8 mA ≈ **450 Ω**. This matches the knee in the plot. Below 450 Ω the zener is off, and $V_o = 20 \times R_L/(1000 + R_L)$: at 100 Ω that is 1.82 V ✓.

WRITE IN YOUR RECORD

Result: % line regulation = **2.8 %** (14–17 V, R_L = 1 kΩ); % load regulation = **0.81 %** (no load → 1 kΩ, V_{in} = 20 V). Lower % = better regulation.

Theory for the record: a zener shunt regulator places a reverse-biased zener across the load, fed through a series resistor R_S. In breakdown, the zener voltage is nearly constant, so changes in input voltage appear across R_S, and changes in load current are absorbed by a change in zener current. Regulation holds while $I_{Z(min)} < I_Z < I_{Z(max)}$. R_S is chosen so that $I_Z \geq I_{Z(min)}$ at minimum input and maximum load. Line regulation = $\Delta V_o/\Delta V_{in} \times 100$ %; load regulation = $(V_{NL} - V_{FL})/V_{FL} \times 100$ %.

TRAP: AN "UNREGULATED" LINE PLOT

If V_o keeps rising with slope ½ all the way to 20 V, the zener is in backwards (forward-biased, clamping at about 0.7 V) or is the wrong part. If V_o sticks at about 0.7 V, it is definitely reversed. The cathode (bar) must face the output node.

CHECK 1

(a) Why does V_o in the line plot still creep up slightly (6.23 → 6.36 V) instead of staying perfectly flat? (b) With V_{in} = 20 V and R_L = 1 kΩ, how much current does the zener carry? (c) What happens to I_Z when the load is removed, and why is that the worst case for zener power?

Answers

Opening question. Across R1. The extra input raises the current through R1 (it flows on into the zener), so R1's voltage drop rises by about 2.92 V.

Check 1. (a) The zener has a dynamic resistance r_z (Exp 1). More current through it means slightly more voltage: $\Delta V = r_z \Delta I_Z$. (b) $I_{R1} = (20 - 6.358)/1000 = 13.64$ mA, $I_L = 6.358/1000 = 6.36$ mA, so $I_Z = 13.64 - 6.36 =$ **7.28 mA**. (c) All of R1's current then flows through the zener, its maximum, so I_Z and $P_Z = V_Z I_Z$ peak. Design must keep this below $I_{Z(max)}$.